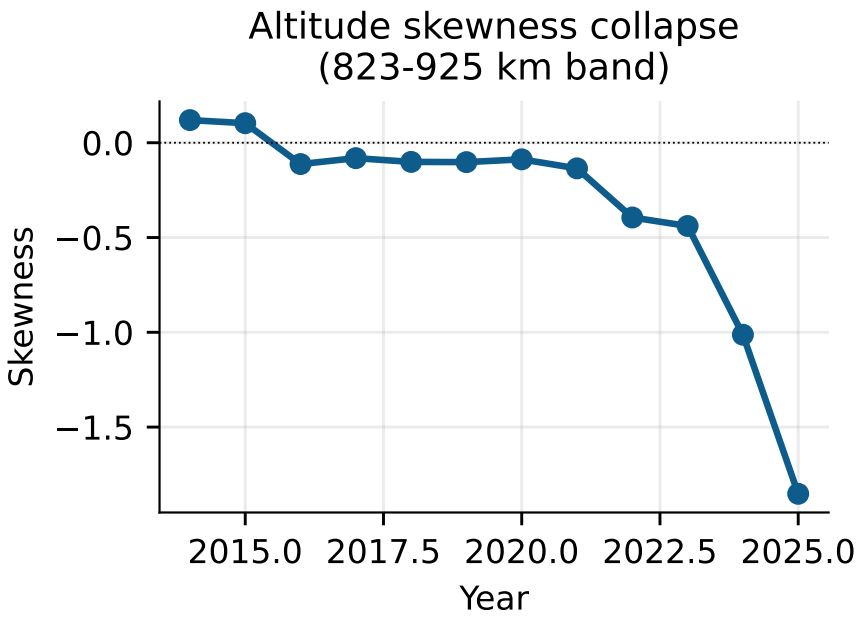
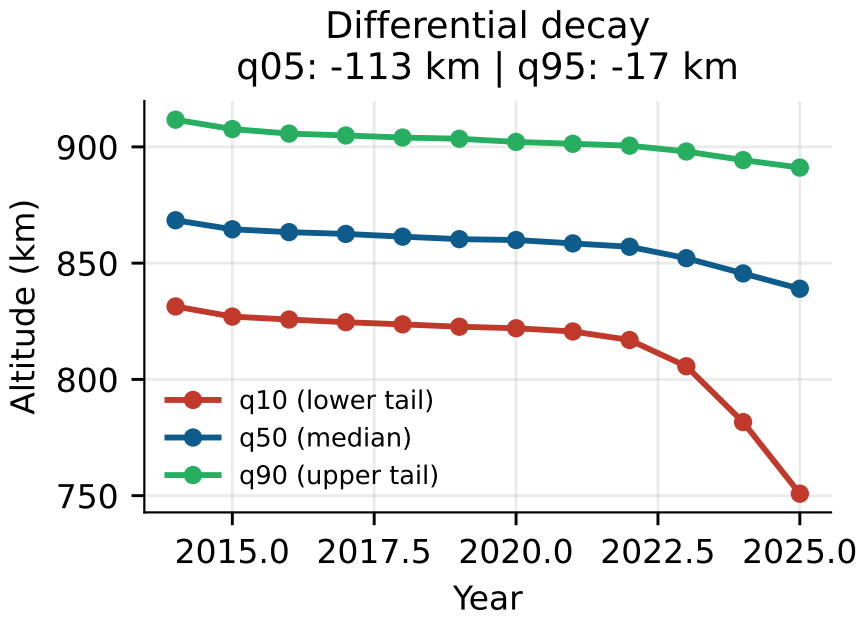
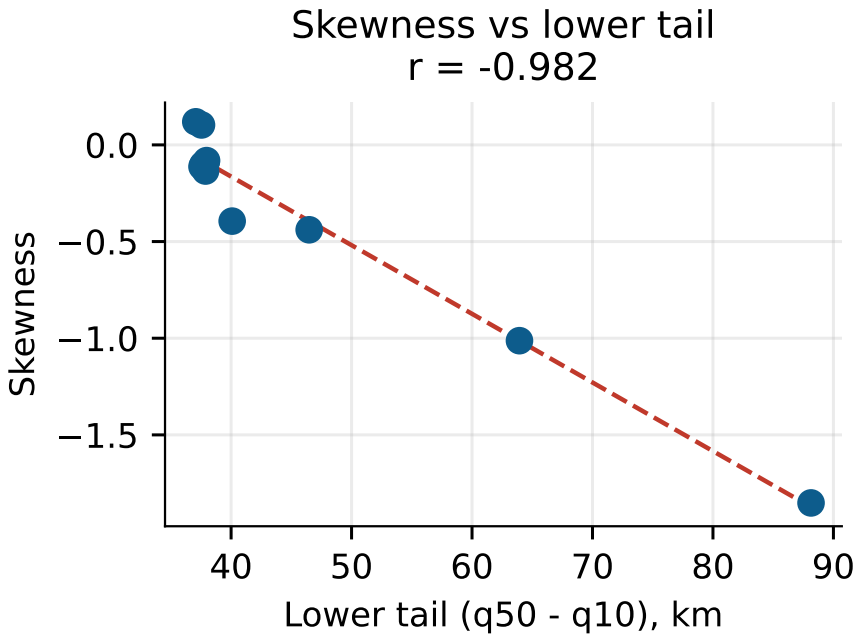
A TLE-based probe of thermospheric density and geomagnetic coupling

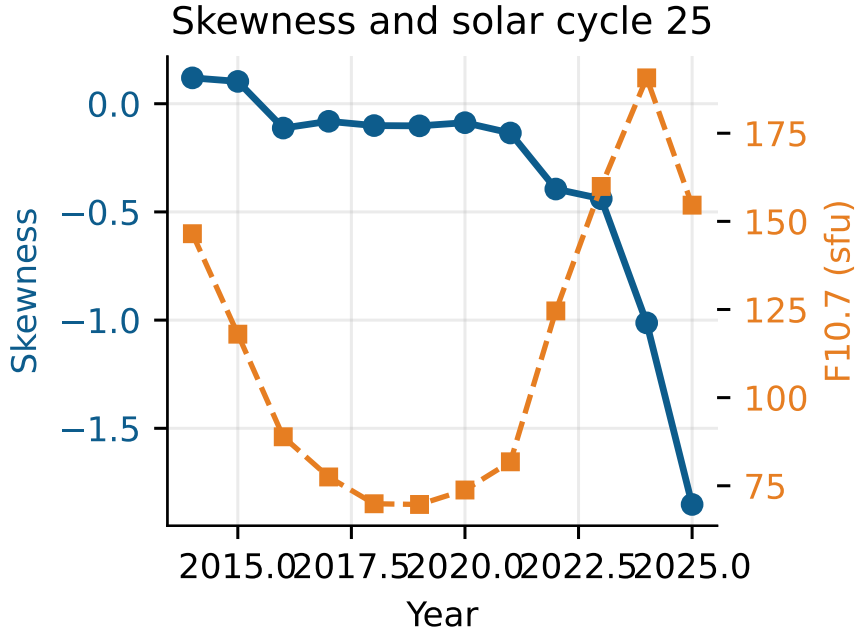
Natanael A. N. Junior | natanaelanjr@gmail.com | Independent Researcher

https://github.com/Nat699/thermoLEO | DOI: 10.5281/zenodo.23376365


Altitude skewness collapse
(823-925 km band)
Skewness
0.0
−0.5
−1.0
−1.5
2015.0
2017.5
2020.0
2022.5
2025.0
Year


Differential decay
q05: -113 km | q95: -17 km
Altitude (km)
900
850
800
750
q10 (lower tail)
q50 (median)
q90 (upper tail)
2015.0
2017.5
2020.0
2022.5
2025.0
Year


Skewness vs lower tail
r = -0.982
Skewness
0.0
−0.5
−1.0
−1.5
40
50
60
70
80
90
Lower tail (q50 - q10), km


Skewness and solar cycle 25
Skewness
0.0
−0.5
−1.0
−1.5
F10.7 (sfu)
175
150
125
100
75
2015.0
2017.5
2020.0
2022.5
2025.0
Year

PROBLEM

The upper thermosphere (800–1000 km) is poorly sampled by dedicated missions (CHAMP, GRACE, Swarm operate below 500 km). Empirical models rely on sparse assimilation and lack independent validation at these altitudes. The coupling between geomagnetic field geometry and thermospheric density in this altitude range is essentially unconstrained observationally.

METHOD

We extract a density signal from the lower tail of the LEO satellite altitude distribution using only public TLE data. A fixed population of 4,108 Sun-synchronous satellites (1,369 in the 823–925 km band) is analyzed yearly from 2014 to 2025. Multi-band tomography covers 11 altitude bands (550–1550 km). Cross-validation uses Swarm in-situ B field and the CHAOS-7 geomagnetic model.

KEY RESULTS

- Skewness collapse: $+0.12$ (2014) $\rightarrow$ -1.85 (2025), slope $= -0.125$/yr ($p = 0.002$)
- Lower tail (q05) decays $6.6\times$ faster than upper tail (q95): -113 km vs -17 km
- **Driver is the radial component** of the geomagnetic field (B_C): Swarm in-situ $r = -0.67$ ($p = 0.0003$), CHAOS-7 $r = -0.61$ ($p = 0.0015$)
- Magnitude $|B|$ does *not* correlate ($r = -0.21$, n.s.), isolating vertical M-I-T coupling as the physical driver
- Orbital-class fingerprint: SSO $+1.37$, polar -0.55, inclined -0.34, equatorial $+0.35$
- Strongest collapse at 1150 km (-2.34); inversion at 550 km ($+0.88$)

APPLICATIONS

- Global thermospheric density monitoring without dedicated instrumentation
- Satellite drag forecasting and reentry prediction (6–12 months ahead)
- Space weather nowcasting from public TLE archives (since 1957)
- Independent validation of JB2008 / NRLMSISE-00 at 800–1000 km
- Operational metric for magnetosphere–ionosphere–thermosphere energy transfer